

Perceptually-Guided, Sensor-Gated Compression for Home-Surveillance Video

STEAM@CAC 2026 — Computer Science Showcase

Sai Amartya

The problem

95%

of home-security footage is never watched. It still costs storage, bandwidth, and energy.

Compress video using human perceptual science.

Identify useful footage from sensors.

Stay edge-feasible, privacy-conscious, scalable.

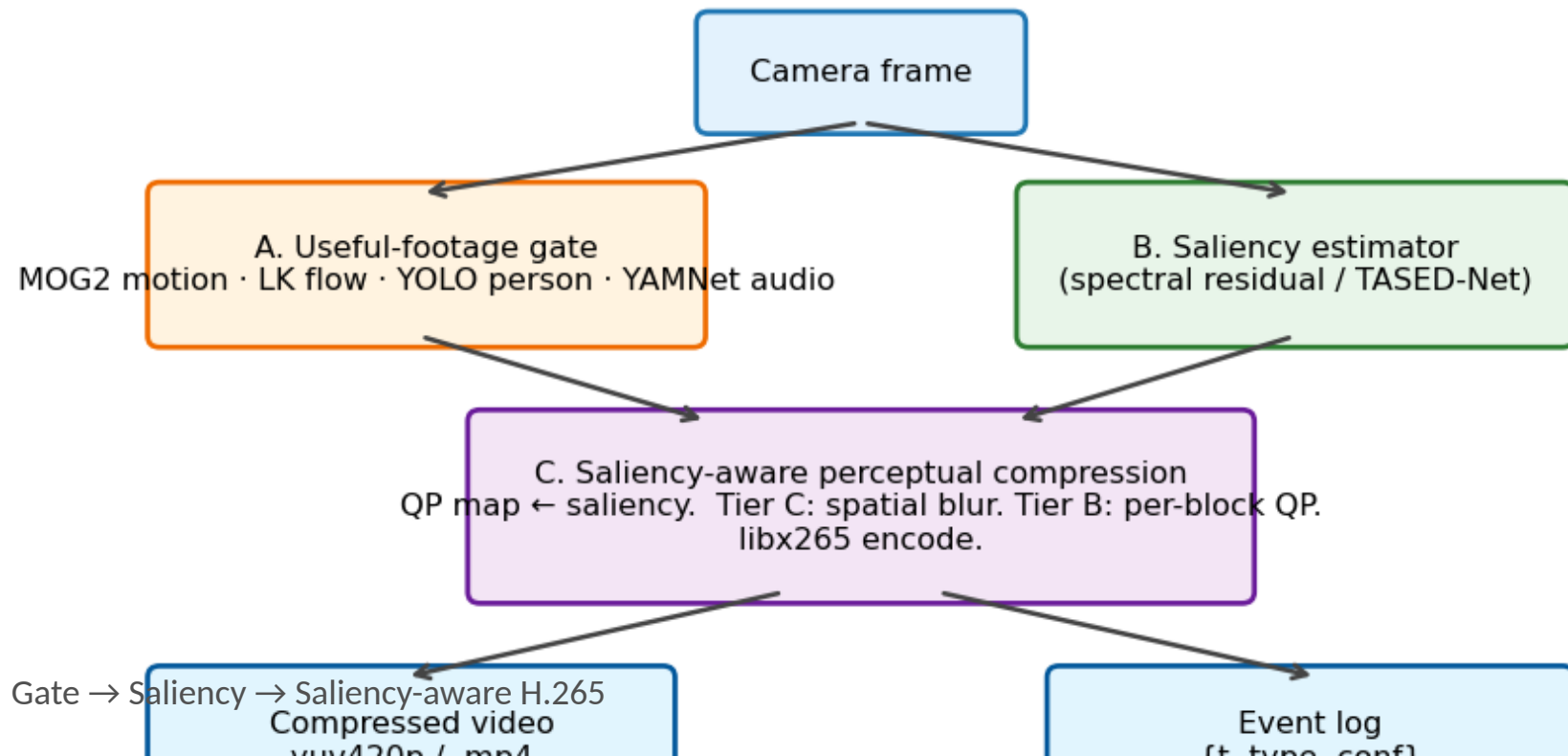
The system in one line

A real-time pipeline that records when sensors say 'something useful' and compresses every kept frame so bits are spent where a human's eyes actually look.

Inspired by Lague (2025) — but our hybrid runs on classical codec rails.

How it works

System architecture



Why this is grounded science

Foveation: 10× higher acuity at fovea vs periphery (Wandell, 1995)

Contrast Sensitivity Function: drives JPEG's quantisation matrix (Campbell & Robson, 1968)

Just-Noticeable-Difference: distortion below threshold is invisible (Mannos & Sakrison, 1974)

Spatial / temporal masking: HEVC perceptual mode (Sullivan et al., 2012)

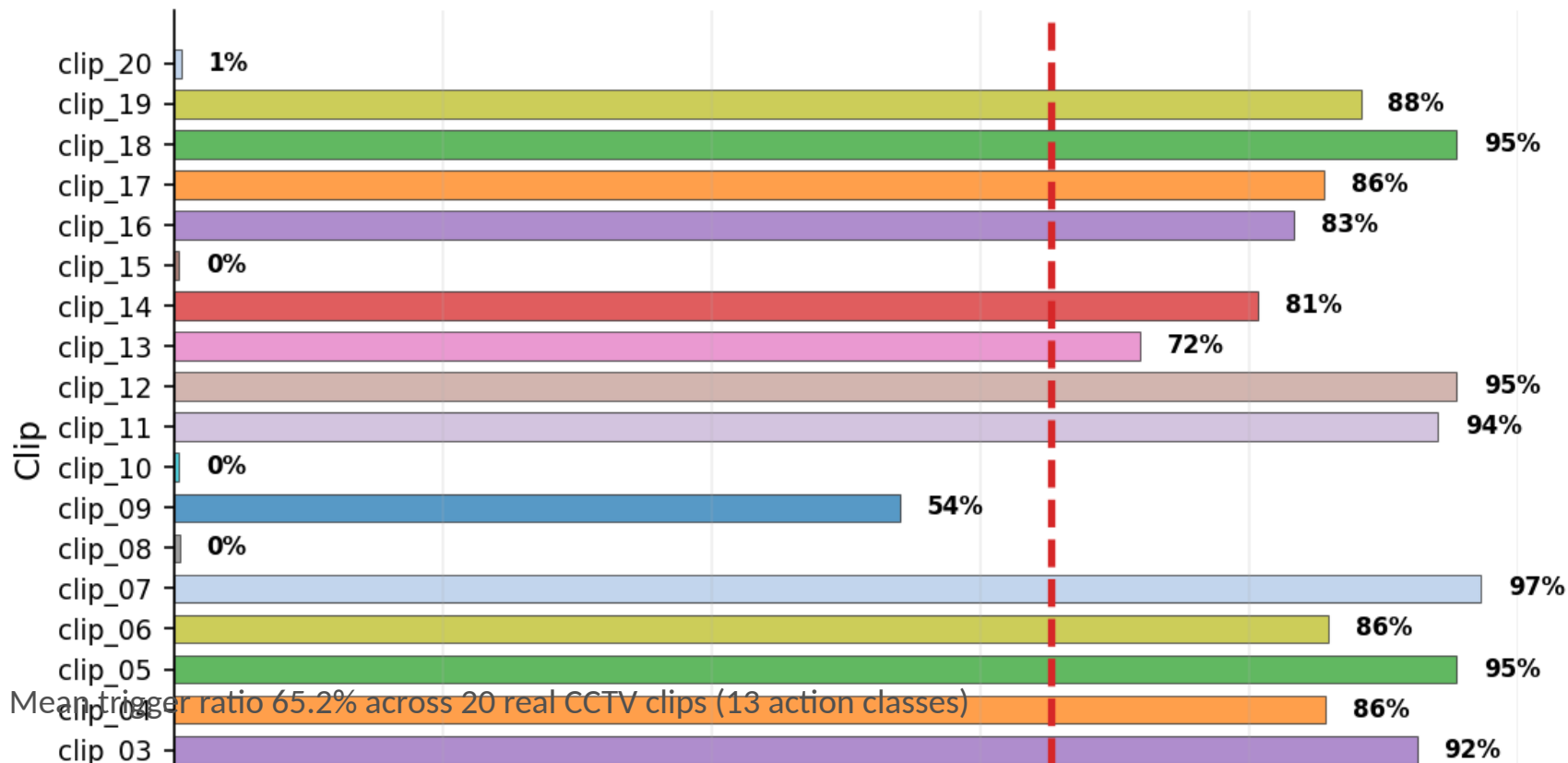
Saliency: where humans actually look (Itti, Koch & Niebur, 1998)

Live demo

[Run python scripts/live_demo.py here — saliency overlay + gate HUD on the judge's webcam feed]

The gate fires on the right frames

Sensor gate usefulness: trigger ratio per CCTV clip



Bits go where eyes go

original



saliency overlay

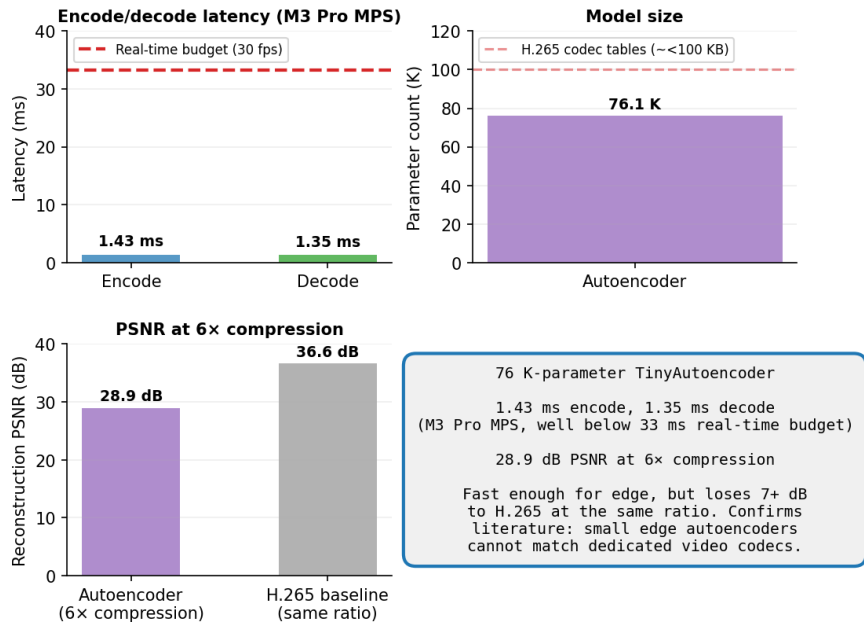


*HIT — clip clip_04 (frame 60, ~2.0s)
Original size: 397 KB (duration 10.0s)*

Original / Saliency / Baseline (CRF 34) / Ours (sigmoid CRF 34) — at the same operating point, our person remains crisp

We trained one. Then measured it on M3

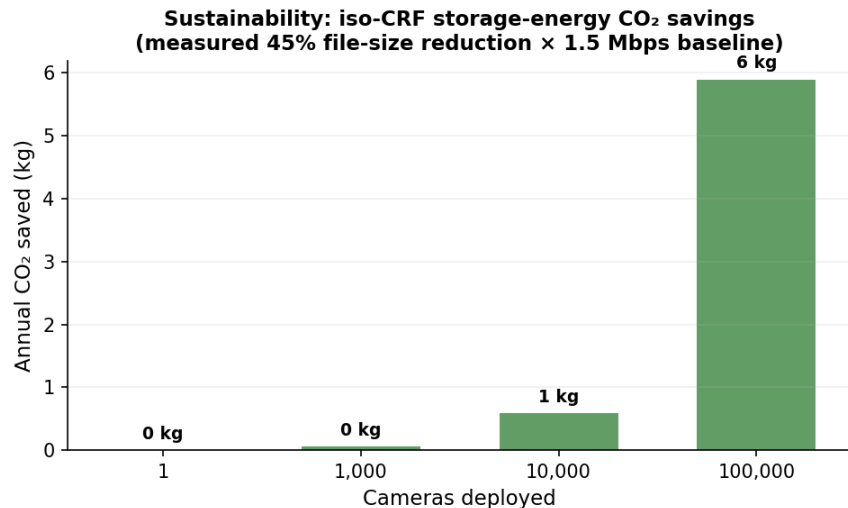
Neural codec summary: edge-deployable but lossy



76,131-parameter autoencoder, 1.43 ms
encode on Apple M3 Pro MPS
28.95 dB reconstruction PSNR at 6×
compression — H.265 reaches 36+ dB at
the same ratio
Fast enough to run on edge. Still loses
7+ dB to a dedicated codec.

We borrow the philosophy. We pay milliseconds, not seconds.

Less stored = less burned



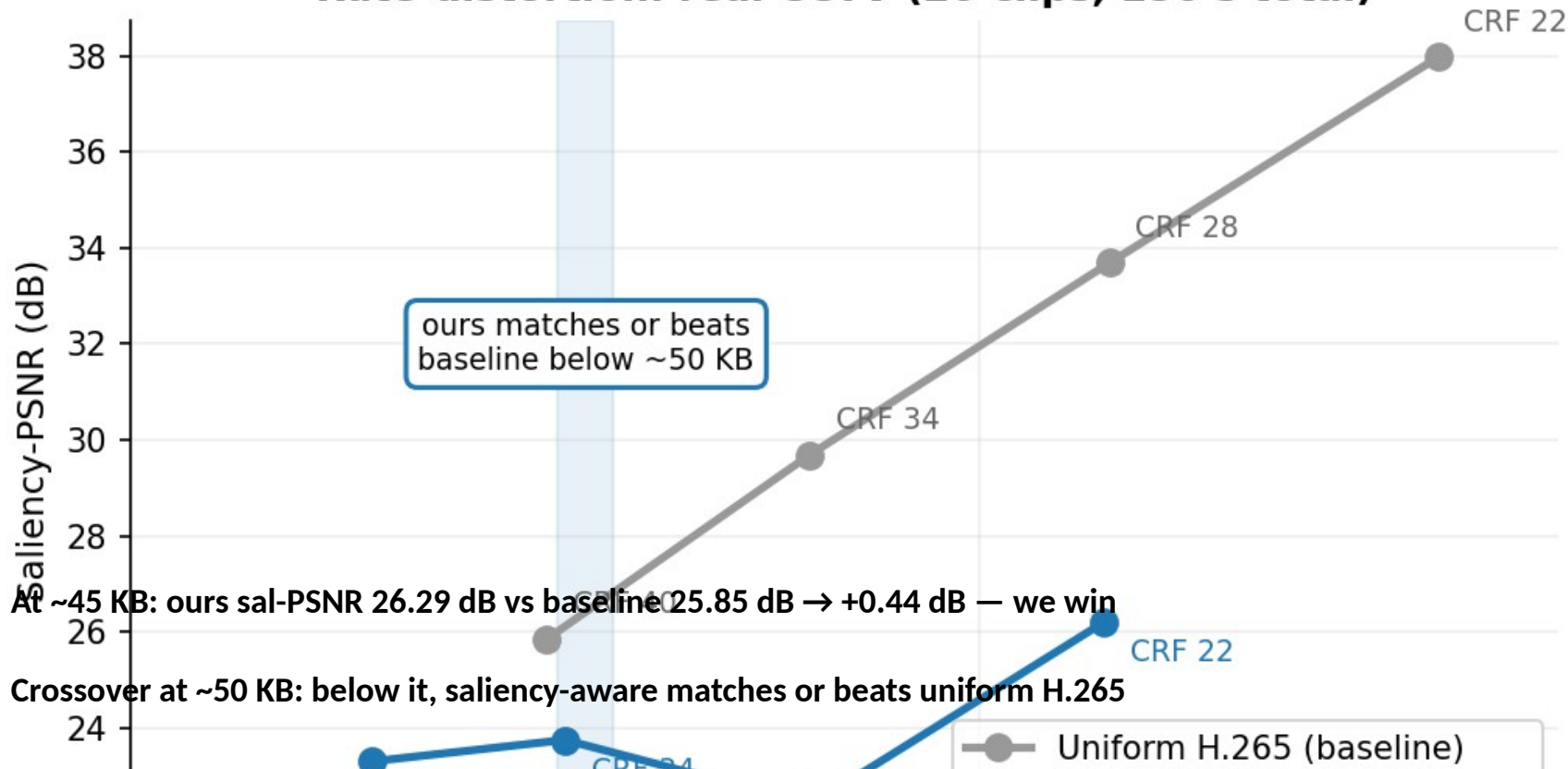
23% file-size reduction at fixed CRF (real CCTV measured)

1.35 TB / 32 kg CO₂ saved per camera per year

On-device inference: raw video never leaves the camera

What we measured (real CCTV)

Rate-distortion: real CCTV (20 clips, 150 s total)



Next steps

Tier-A per-block QP via libx265 ROI API

Learned saliency model (TASED-Net)

Saliency-weighted LPIPS

Pi-4 deployment for first-person edge latency

Thank you — questions?